Certified Finite-Trial Approximation for the Mézard–Parisi Diluted Spin-Glass Functional

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Abstract

The supplied September 23 manuscript claims the Mézard–Parisi variational equality for an even-arity, first-moment class of diluted Ising models. This report studies certified evaluation and approximation of its actual hierarchical trial functional, not a substitute replica-symmetric functional. A spin-contrast modulus controls leaf-message clipping and finite-alphabet replacement uniformly over every finite depth and every admissible mass vector, without exponential disorder moments. Exact removal of the interaction midpoint then gives an omitted-degree Poisson budget rather than a whole-bad-event budget. A normalized moment evaluation of conditional power means avoids an artificial inverse-mass singularity, with explicit accumulated errors and positivity gates. A one-level two-spin example supplies an independently reproducible enclosure. The source's all-multioverlap decorrelation and limiting cavity comparison remain global obligations, not consequences of these finite certificates. Contribution assessment: C3/5, the depth-uniform contrast approximation and assembled finite-trial error bounds are notable local refinements, without an independent proof of the variational equality or a historical novelty claim.

1 Model, conventions, and claim boundaries

Fix an even integer $p \geq 2$ and $\alpha > 0$. A random interaction $\theta : \{-1,1\}^p \to \mathbb{R}$ and an external field h satisfy

$$B(\theta) = \max_s |\theta(s)|, \qquad \mathbb{E}B(\theta) + \mathbb{E}|h| < \infty. \tag{1.1}$$

There are Poisson(αN) independent interactions on independently uniform ordered p-tuples of sites, with replacement. Independent copies of h act at the sites. With $-H_N$ the sum of these interactions and fields, the source uses the counting, not probability, spin prior:

$$F_N = N^{-1}\mathbb{E}\log \sum_{\sigma\in\{-1,1\}^N} e^{-H_N(\sigma)}. \tag{1.2}$$

Its upper-bound class additionally requires

$$e^{\theta(s)} = a\left(1 + b\prod_{\ell=1}^{p} f_\ell(s_\ell)\right), \quad a > 0, \qquad \mathbb{E}(-b)^n \geq 0 \quad (n \geq 1), \tag{1.3}$$

where the f_ℓ are identically distributed independent random functions, independent of b, and $|b\prod_\ell f_\ell(s_\ell)| < 1$ almost surely for every spin vector. The moments in (1.3) are finite. The factor a may depend on the other data; no integrability of $\log a$ separately is assumed. Evenness of p is used in the source's convex interpolation remainder. None of the local evaluation lemmas below supplies the missing sign for odd p or a formula for random 3-SAT.

For $x \in \mathbb{R}$, let $q_x(s) = e^{xs}/(2\cosh x)$, with mean $\tanh x$. Set

$$C_\theta(x_1, \dots, x_p) = \sum_{s\in\{-1,1\}^p} e^{\theta(s)} \prod_{\ell=1}^{p} q_{x_\ell}(s_\ell), \tag{1.4}$$

$$U_\theta(x_1, \dots, x_{p-1}; \epsilon) = \log \sum_{s\in\{-1,1\}^{p-1}} e^{\theta(s,\epsilon)} \prod_{\ell<p} q_{x_\ell}(s_\ell). \tag{1.5}$$

In particular $|\log C_\theta|, |U_\theta| \le B(\theta)$ regardless of the message size.

Write $\mathcal{M}_1 = \mathcal{P}(\mathbb{R})$ and $\mathcal{M}_{j+1} = \mathcal{P}(\mathcal{M}_j)$, with their weak Borel structures. For depth $r \ge 1$, choose $\zeta \in \mathcal{M}_{r+1}$ and $0 < m_1 < \cdots < m_r < 1$. Each message label independently draws

$$\eta_0 \sim \zeta, \qquad \eta_{d+1} \mid \eta_d \sim \eta_d \ (0 \le d < r-1), \qquad x \mid \eta_{r-1} \sim \eta_{r-1}.$$

The *whole chains*, not only the terminal messages, are independent between labels and independent of all physical disorder. For $K \sim \mathrm{Poisson}(\lambda)$, $\lambda = \alpha p$, let

$$V_s = \frac{1}{2}\sum_{\epsilon=\pm1} \exp\left(h\epsilon + \sum_{j=1}^{K} U_{\theta_j}(x_1^j, \dots, x_{p-1}^j; \epsilon)\right), \qquad V_e = C_\theta(x_1, \dots, x_p). \tag{1.6}$$

All labels in these formulas are distinct. Let $\mathscr{F}_d$ contain all physical disorder and all label variables η_j for $j \le d$. The recursion and trial functional are

$$T_r V = V, \qquad T_d V = \left(\mathbb{E}[(T_{d+1}V)^{m_{d+1}} \mid \mathscr{F}_d]\right)^{1/m_{d+1}}, \tag{1.7}$$

$$\mathcal{B}_r = \log 2 + \mathbb{E}\log T_0 V_s - \alpha(p-1)\mathbb{E}\log T_0 V_e. \tag{1.8}$$

Only auxiliary variables are integrated inside T_d; h, K, θ_j remain quenched. At $r = 0$, take iid messages from $\zeta \in \mathcal{P}(\mathbb{R})$ and omit the recursion. Then

$$|\log T_0 V_s| \le |h| + \sum_{j=1}^{K} B(\theta_j), \qquad |\log T_0 V_e| \le B(\theta). \tag{1.9}$$

These ensure finite trial values with no message moments.

The supplied manuscript [1] claims $\lim_N F_N = \inf_{r\ge0,\zeta,\mathbf{m}} \mathcal{B}_r$. We refer to this as the *source theorem*, not an independently certified conclusion of this report. Our local results apply to specified trial laws and their infima as mathematical functionals. The classical upper bound $F_N \le \mathcal{B}_r$ is used only where explicitly identified.

2 Exact centering and a contrast modulus

Lemma 2.1 (Conditional logarithmic contraction)**.** *For a probability kernel P and $m > 0$, put $\mathcal{L}_{m,P}(Z) = m^{-1}\log Pe^{mZ}$. If $|Z - Z'| \le D$, where D is measurable before applying the kernel, then $|\mathcal{L}_{m,P}(Z) - \mathcal{L}_{m,P}(Z')| \le D$. Also $\mathcal{L}_{m,P}(Z+c) = \mathcal{L}_{m,P}(Z) + c$ for such c. Both statements propagate through any finite tower.*

Proof. Multiply $e^{mZ'}$ by $e^{\pm mD}$ to bound e^{mZ}, integrate, take logarithms, and divide by m. Measurability permits the factors to leave the conditional expectation. Apply the same argument at each remaining level. No factor r or m^{-1} is lost. □

Define the midpoint and radius

$$c_\theta = \tfrac{1}{2}(\max_s \theta(s) + \min_s \theta(s)), \qquad R_\theta = \tfrac{1}{2}(\max_s \theta(s) - \min_s \theta(s)), \qquad \bar{\theta} = \theta - c_\theta. \tag{2.1}$$

Both $|c_\theta|$ and R_θ are bounded by $B(\theta)$.

Proposition 2.2 (Removal of common interaction offsets)**.** *For every trial, with the same messages, field, and Poisson law,*

$$\mathcal{B}_r(\theta, h) = \alpha \mathbb{E} c_\theta + \mathcal{B}_r(\bar{\theta}, h). \tag{2.2}$$

The midpoint uniquely minimizes $\max_s |\theta(s) - c|$ *over* $c \in \mathbb{R}$*, with minimum* R_θ*.*

Proof. Each site interaction contributes the additive c_{θ_j} to U, independent of ϵ and of all messages. Lemma 2.1 extracts $\sum_{j \le K} c_{\theta_j}$ from the site logarithm and c_θ from the edge logarithm. Taking expectations gives $(\alpha p - \alpha(p-1))\mathbb{E} c_\theta = \alpha \mathbb{E} c_\theta$. For endpoints $u = \min \theta$, $v = \max \theta$, the quantity to minimize is $\max(v - c, c - u)$, which is at least $(v-u)/2$, with equality exactly at $(u+v)/2$. This proof also covers $u = v$. □

The source already proves extraction of any integrable common shift. Our choice is its minimax specialization for the following error budgets, not a newly discovered invariance. It does not separate $\log a$ in (1.3). A large random common offset can have an integrable mean and still dominate B; it should not inflate message errors.

We use $\mathrm{TV}(P, Q) = \frac{1}{2}\|P - Q\|_1$ on a finite space.

Lemma 2.3 (Positive contrast bound)**.** *If* $e^{-R} \le f \le e^R$ *and* $\mathrm{TV}(P, Q) \le t \in [0, 1]$*, then*

$$|\log Pf - \log Qf| \le g(R, t) := \log(1 + t(e^{2R} - 1)) \le 2R. \tag{2.3}$$

For product measures with q *factors and each factor within total variation* ε*, their total variation is at most* $t_q(\varepsilon) = 1 - (1 - \varepsilon)^q$*.*

Proof. Since $P - Q$ has total mass zero, $|(P - Q)f| \le (e^R - e^{-R})t$. Both integrals are at least e^{-R}. Divide their positive difference by the smaller integral to obtain (2.3). For the product statement, couple each pair of factors maximally and independently. The probability of at least one mismatch is at most $1 - (1 - \varepsilon)^q$, which bounds total variation. □

3 Depth-uniform message approximation

Let $Q : \mathbb{R} \to \mathbb{R}$ be Borel and

$$\sup_x \mathrm{TV}(q_x, q_{Q(x)}) = \tfrac{1}{2} \sup_x |\tanh x - \tanh Q(x)| \le \varepsilon \le 1. \tag{3.1}$$

Push Q through a hierarchy recursively: push a leaf measure forward by Q, then push a measure on leaf measures by that induced map, and continue to ζ. This retains the conditional law of every prefix and couples original and transformed terminal variables. It is not replacement of the unconditional leaf marginal alone.

Theorem 3.1 (Uniform contrast certificate)**.** *Under (1.1), for every finite r, every admissible mass vector, and every trial law, the transformed trial satisfies*

$$|\mathcal{B}_r - \mathcal{B}_r^Q| \le \mathcal{E}(\varepsilon), \tag{3.2}$$

$$\mathcal{E}(\varepsilon) := \alpha p\, \mathbb{E}g(R_\theta, t_{p-1}(\varepsilon)) + \alpha(p-1)\, \mathbb{E}g(R_\theta, t_p(\varepsilon)). \tag{3.3}$$

In particular $\mathcal{E}(\varepsilon) \to 0$ as $\varepsilon \downarrow 0$, uniformly over these depths, masses, and laws. No exponential moment of R_θ is required.

Proof. Subtracting c_θ leaves both logarithmic differences unchanged. Apply Lemma 2.3 to $e^{\bar\theta}$ with p product factors for the edge, and with $p-1$ factors for each fixed last spin in U. For the site, the difference of logarithms of the two positive spin sums is at most $\sum_{j\le K} g(R_{\theta_j}, t_{p-1})$. This bound is measurable with respect to the physical root disorder. It therefore propagates through the entire tower by Lemma 2.1; so does $g(R_\theta, t_p)$ for the edge. The recursively pushed-forward hierarchy can be evaluated on the original latent chain: conditional sampling followed by a pushforward equals sampling its pushforward. Induction through the kernels proves that this evaluation is exactly $\mathcal{B}_r^Q$, even if Q merges labels' possible values. Independent chains remain independent. Finally $\mathbb{E}\sum_{j\le K} g(R_{\theta_j}, t) = \alpha p\, \mathbb{E}g(R_\theta, t)$. Dominated convergence uses $0 \le g \le 2R_\theta$ and $\mathbb{E}R_\theta < \infty$, not a bound on $\mathbb{E}e^{2R_\theta}$. □

Corollary 3.2 (Clipping and finite alphabets)**.** *For $Q(x) = \mathrm{clip}_{[-L,L]}\, x$, one can take $\varepsilon_L = (1 + e^{2L})^{-1}$. For every $\varepsilon > 0$ there is a finite set $G_\varepsilon \subset \mathbb{R}$ and a Borel Q with range in G_ε satisfying (3.1). If I is the infimum of the original trial functional over all finite depths and I_ε the same infimum restricted to these terminal alphabets, then*

$$I \le I_\varepsilon \le I + \mathcal{E}(\varepsilon). \tag{3.4}$$

Proof. For clipping, the supremum difference of spin means is $1 - \tanh L = 2/(1 + e^{2L})$. Cover $[-1, 1]$ by finitely many intervals of radius 2ε with centers in $(-1, 1)$; a measurable nearest-center choice followed by arctanh gives Q. The restricted infimum is over a subset. For the other inequality, push forward a trial within any $\eta > 0$ of I and use Theorem 3.1; then let $\eta \downarrow 0$. Finiteness and a uniform lower bound for these infima follow from (1.9). □

The hierarchy measures above a finite alphabet may still have continuous support. Thus (3.4) is not a finite-branching optimization algorithm, an attainment theorem, or a computable lower bound on I. Nor does it establish $I = \lim F_N$.

Proposition 3.3 (First-moment tail modulus)**.** *For $A \ge 0$, $0 \le t \le 1$,*

$$\mathbb{E}g(R, t) \le \log(1 + t(e^{2A} - 1)) + 2\mathbb{E}(R - A)_+. \tag{3.5}$$

For message clipping and $q \ge 1$, this is at most $\log(1 + qe^{2(A-L)}) + 2\mathbb{E}(R - A)_+$.

Proof. The derivative $\partial_R g = 2te^{2R}/(1 - t + te^{2R})$ lies in $[0, 2]$. Thus $g(R, t) \le g(A, t) + 2(R - A)_+$; take expectations. Also $t_q(\varepsilon_L) \le q\varepsilon_L \le qe^{-2L}$. □

If $R \le A_0$, take $A = A_0$ for an exponentially small clipping bound. If $\mathbb{P}(R > u) = (1 + u)^{-\gamma}$ with $\gamma > 1$, then $\mathbb{E}(R - A)_+ = (1 + A)^{1-\gamma}/(\gamma - 1)$; choosing $A = L/2$ gives an explicit algebraic tail rate plus $O(e^{-L})$. First-moment existence alone gives convergence but no uniform numerical rate over all such disorder laws. Rare, arbitrarily large contrasts can defeat any rate not accompanied by a tail certificate.

4 Poisson capping and a combined physical budget

After (2.2), keep $\alpha\mathbb{E}c_\theta$ exactly outside the trial. For an integer $J \geq 0$, replace K only in the centered site term by $K_J = \min(K, J)$, retaining the first K_J interactions and their whole label chains. The edge term is unchanged. Denote the result by $\mathcal{B}_{r,J}(\bar{\theta}, h)$. It is a defined approximation, not generally a member of the original Poisson trial family.

Theorem 4.1 (Omitted-degree rather than bad-event cost)**.** *For any hierarchy,*

$$\left|\mathcal{B}_r(\theta, h) - \left(\alpha\mathbb{E}c_\theta + \mathcal{B}_{r,J}(\bar{\theta}, h)\right)\right| \leq \mathbb{E}R_\theta\, \tau_J(\lambda), \qquad \tau_J(\lambda) = \mathbb{E}(K - J)_+. \tag{4.1}$$

For $K \sim \mathrm{Poisson}(\lambda)$,

$$\tau_J(\lambda) = \lambda\mathbb{P}(K \geq J) - J\mathbb{P}(K \geq J + 1). \tag{4.2}$$

Proof. Deleting one centered interaction changes each exponent in the two-spin sum by at most R_{θ_j}. Deleting all omitted interactions changes the site logarithm by at most $\sum_{j=J+1}^{K} R_{\theta_j}$. This physical-disorder bound passes through every conditional kernel. Unused independent labels integrate to one and do not alter the retained hierarchy. Independence of K and the interactions gives (4.1). For (4.2), use $k\mathbb{P}(K = k) = \lambda\mathbb{P}(K = k - 1)$ in $\sum_{k \geq J+1}(k - J)\mathbb{P}(K = k)$. All summands are nonnegative and integrable, justifying reindexing. □

The cost contains neither $|h|\mathbb{P}(K > J)$ nor the cost of throwing away the first J interactions on that event. This improves the source comparison proof's whole-event envelope when used for trial evaluation. It does not change the source's limiting decorrelation argument or produce its finite-N error rate. Extracting the offset *before* capping is important: capping an uncentered interaction would otherwise charge its common shift with coefficient $\mathbb{E}K_J$, where the exact cancellation requires αp.

For a practical upper bound on (4.2), let $p_j = e^{-\lambda}\lambda^j/j!$. If $J + 2 > \lambda$, then

$$\tau_J(\lambda) = p_{J+1}\sum_{\ell \geq 0}(\ell + 1)\frac{\lambda^\ell(J + 1)!}{(J + 1 + \ell)!} \leq \frac{p_{J+1}}{(1 - \lambda/(J + 2))^2}. \tag{4.3}$$

Indeed each successive factorial ratio is at most $(\lambda/(J + 2))^\ell$, and $\sum_{\ell \geq 0}(\ell + 1)z^\ell = (1 - z)^{-2}$. The exact expression (4.2) may be sharper; interval arithmetic should account for cancellation if it is used numerically.

Theorem 4.2 (Assembled truncation certificate)**.** *Fix a trial and a transform Q as in (3.1). For $A, H \geq 0$ define $\theta_A(s) = \mathrm{clip}_{[-A,A]}\,\bar{\theta}(s)$ and $h_H = \mathrm{clip}_{[-H,H]}\,h$. Let $\widehat{B}$ be an evaluation of the transformed, degree-capped trial $\alpha\mathbb{E}c_\theta + \mathcal{B}^Q_{r,J}(\theta_A, h_H)$ with certified absolute evaluation error at most ϵ_{ev}. If $\alpha\mathbb{E}c_\theta$ is itself only approximated, include its absolute error in ϵ_{ev}. Put $\lambda_J = \mathbb{E}K_J$. Then*

$$\begin{aligned}|\mathcal{B}_r(\theta, h) - \widehat{B}| \leq\ & \mathcal{E}(\varepsilon) + \mathbb{E}R_\theta\, \tau_J(\lambda) + \mathbb{E}(|h| - H)_+ \\ & + [\lambda_J + \alpha(p - 1)]\mathbb{E}(R_\theta - A)_+ + \epsilon_{\mathrm{ev}}.\end{aligned} \tag{4.4}$$

The bound is uniform in the trial depth and masses except for whatever depth dependence is explicitly present in ϵ_{ev}.

Proof. Apply Theorem 3.1, then the exact offset extraction and Theorem 4.1. At each spin vector $|\bar{\theta} - \theta_A| \leq (R_\theta - A)_+$. Positivity of all spin averages bounds a site-message change by this number and an edge logarithm change by the same number. The retained site has expected degree λ_J; the edge coefficient is $\alpha(p - 1)$. The external-field change is bounded by $(|h| - H)_+$ inside the site logarithm. Lemma 2.1 passes each bound through the hierarchy, and the triangle inequality then proves (4.4). □

This ordering of operations avoids double charging removed interactions. The clipping of interaction entries need not retain (1.3). Consequently one must not apply a new interpolation upper bound to the clipped model merely because it is convenient to evaluate. Instead, if the original model's upper bound is available, (4.4) gives $F_N \le \widehat{B} +$ right-hand error. A lower enclosure of one trial value is *not* a lower bound on F_N.

5 Normalized conditional moment evaluation

The following is a local, deterministic use of the finite-remainder method in the original Ram moment chapter [8]. It is not a claim that finitely many raw message moments determine a hierarchical trial. At a specified prefix, keep the physical disorder fixed. Suppose the continuation logarithm Z has a certified range $a \le Z \le a + W$. For $0 < m < 1$, define

$$\phi_m(z) = \frac{e^{mz} - 1}{m}, \quad \phi_0(z) = z, \qquad \mathcal{L}_{m,P}(Z) = a + \frac{\log(1 + mP\phi_m(Z - a))}{m}. \tag{5.1}$$

The subtraction of the exact constant occurs *before* approximating the expectation.

Theorem 5.1 (A mass-stable conditional certificate)**.** *Let P be the actual conditional probability law and let Q be a positive normalized quadrature supported where $0 \le Z - a \le W$. If $|P\phi_m(Z - a) - Q\phi_m(Z - a)| \le \eta$, then the approximation in (5.1) with P replaced by Q differs by at most η, uniformly as $m \downarrow 0$. For $q \ge 1$, if $|P(Z - a)^j - Q(Z - a)^j| \le e_j$ for $1 \le j \le q$, one can take*

$$\eta = \sum_{j=1}^{q} \frac{m^{j-1} e_j}{j!} + \frac{2m^q e^{mW} W^{q+1}}{(q + 1)!}. \tag{5.2}$$

Proof. Both expectations of ϕ_m are nonnegative. The function $t \mapsto m^{-1} \log(1 + mt)$ has derivative $(1 + mt)^{-1} \le 1$ on $[0, \infty)$. The mean value theorem proves the first assertion. Taylor's integral remainder on $[0, W]$ gives

$$\phi_m(z) = \sum_{j=1}^{q} \frac{m^{j-1} z^j}{j!} + \mathcal{R}_q(z), \qquad 0 \le \mathcal{R}_q(z) \le \frac{m^q e^{mW} W^{q+1}}{(q + 1)!}.$$

Integrate the polynomial error and bound the two residual integrals. Normalization and positivity give each residual integral norm at most the displayed supremum, proving (5.2). □

The factor 2 is safe rather than optimal; nonnegativity of both remainders can sometimes reduce it. The significant feature here is the absence of an artificial $1/m$ amplification. Directly bounding an absolute error in Pe^{mZ} and then dividing by m conceals the cancellation of the exactly integrated constant. As $m \downarrow 0$, the operator tends to PZ and (5.2) tends to the first-moment error e_1. Stable software would use `expm1` and `log1p`, but certified arithmetic, not a software function name, determines ϵ_{ev}.

Proposition 5.2 (Signed rules and accumulation)**.** *If $Q1 = 1$ but Q is signed with total variation Λ, replace the residual factor 2 in (5.2) by $1+\Lambda$. In addition one must certify $1+mQ\phi_m(Z-a) \ge \ell > 0$; the resulting log error is at most $\eta/\min(1, \ell)$. For a finite tower evaluated with* positive normalized *kernels, if each stage has a uniform conditional log error at most η_d, then its root log error is at most $\sum_{d=0}^{r-1} \eta_d$.*

Proof. The signed residual integral is bounded by Λ times its uniform norm, exactly as in the finite cubature proof of [10]. The true logarithm argument is at least 1. The derivative of $\log u$ on the segment joining it to the approximate argument is at most $1/\min(1,\ell)$; the input argument error is $m\eta$, which cancels division by m. For accumulation, compare an exact outer operator on the exact and computed continuation, then the exact and approximate operators on the computed continuation. Lemma 2.1 gives the inherited error unchanged and the stage certificate adds η_d. Induct backwards. A certified range for the computed continuation must be used at every stage; it is not inferred from an unverified numerical value. □

For the capped, physically clipped model, uniform continuation ranges can be taken as $[-(H + JA), H + JA]$ at the site and $[-A, A]$ at the edge. If the corresponding stage errors are $\eta_{s,d}, \eta_{e,d}$, their contribution to (4.4) is

$$\sum_d \eta_{s,d} + \alpha(p-1)\sum_d \eta_{e,d}, \tag{5.3}$$

plus certified errors in the outer physical expectation and the offset. There is no free depth-uniform quadrature accuracy: per-stage budgets must be summable or adjusted to the chosen finite r. The needed moments are conditional moments of the *actual continuation logarithm*, not unconditional moments of x. A finite alphabet at the leaves does not make these higher conditional laws finite.

For a specified finite-branching hierarchy and finite physical support, one can enumerate each conditional state and compute positive weighted power means backwards, enclosing all powers and logarithms. The number of states may grow exponentially with labels and depth. Neither the analysis nor the exact tests assert a polynomial-time algorithm for arbitrary input hierarchies.

5.1 Variation and normalization are inseparable

For a fixed conditional law P and a differentiable bounded path Z_t, direct differentiation gives

$$\frac{d}{dt}\mathcal{L}_{m,P}(Z_t) = P_t Z_t', \qquad P_t(dz) = \frac{e^{mZ_t(z)}P(dz)}{Pe^{mZ_t}}, \qquad \frac{d^2}{dt^2}\mathcal{L}_{m,P}(Z_t) = P_t Z_t'' + m\,\mathrm{Var}_{P_t}(Z_t'). \tag{5.4}$$

The bounds and differentiability are required on a neighborhood of the path to justify differentiation under the integral. The first identity is the quotient-rule calculation; differentiating the normalized weight gives $m(Z_t' - P_t Z_t')$, proving the second. This is classical exponential tilting and already underlies the source tree identities. The original nonlinear Ram proof [9] likewise retains the covariance term in a normalized kernel derivative. If the conditional kernel itself changes, its derivative must also be included; (5.4) is not a formula for an arbitrary variation of ζ with a frozen denominator.

Example 5.3 (Why the hierarchy cannot be flattened)**.** For a one-level positive insertion V taking values 1 and 4, compare two latent constructions. In the first, the root selects δ_1 or δ_4 with equal probability; in the second, the root measure is deterministically $(\delta_1 + \delta_4)/2$. Their unconditional V laws agree. For $m = 1/2$, however,

$$\mathbb{E}\log T_0 V = \log 2 \quad \text{in the first case,} \qquad \mathbb{E}\log T_0 V = 2\log(3/2) = \log(9/4) \quad \text{in the second.}$$

This is a power-mean counterexample, not claimed to be an optimized spin-glass trial. It shows precisely which conditional information a leaf-marginal replacement would discard.

Similarly, physical h cannot be moved into these expectations under only (1.1). For a nonnegative field with $\mathbb{P}(h > t) = (1+t)^{-2}$, $\mathbb{E}h = 1$ but $\mathbb{E}e^{mh} = \infty$ for every $m > 0$. The source's quenched ordering prevents this unnecessary and generally false integrability requirement.

6 An explicit admissible two-spin trial

Take $p = 2$, $\alpha = 1/2$, $h = 0$, and $\theta(s_1, s_2) = Js_1s_2$ with $J = \pm \log 2$ equally likely. Then

$$c_\theta = 0, \quad R_\theta = \log 2, \quad \cosh J = \frac{5}{4}, \quad \tanh J = \pm\frac{3}{5}.$$

The factorization has $a = 5/4$, $b = \tanh J$, and $f_\ell(s) = s$. The sign condition holds because odd moments of b vanish and even ones are nonnegative. Strict positivity holds since $|b| = 3/5 < 1$. Use $r = 1$, $m_1 = 1/2$, and $\zeta = \delta_\nu$, where ν gives probability $1/2$ to each of $x = \pm \operatorname{arctanh}(1/2) = \pm\frac{1}{2}\log 3$. Each label draws its own independent sign.

Cap the site degree at $J_{\mathrm{cap}} = 1$ (distinguished from the coupling J). For zero retained interactions, $V_s = 1$. For one,

$$e^{U_J(x;\epsilon)} = \frac{5}{4}(1 + \tanh J \, \tanh x \, \epsilon), \qquad V_s = \frac{5}{4}.$$

Thus the site contribution is $(1 - e^{-1}) \log(5/4)$, since $K \sim \mathrm{Poisson}(1)$. Conditionally on either physical coupling sign, the edge insertion takes values $17/16$ and $23/16$ equally likely. Consequently

$$T_0 V_e = \left(\frac{\sqrt{17} + \sqrt{23}}{8}\right)^2 = \frac{20 + \sqrt{391}}{32}, \qquad B_{\mathrm{cap}} = \log 2 + (1 - e^{-1}) \log \frac{5}{4} - \frac{1}{2} \log \frac{20 + \sqrt{391}}{32}. \tag{6.1}$$

The uncapped value satisfies the rigorous, explicit enclosure

$$|\mathcal{B}_1 - B_{\mathrm{cap}}| \le e^{-1} \log 2, \qquad F_N \le B_{\mathrm{cap}} + e^{-1} \log 2 \quad \text{if the source-class upper bound is invoked.} \tag{6.2}$$

The first assertion is Theorem 4.1 and $\mathbb{E}(K - 1)_+ = e^{-1}$; the second transfers a trial upper bound. This simple cap is deliberately loose and is not an optimized trial. For a larger cap, one must keep the field-dependent two-spin sum, not raise the one-interaction average to the number of interactions. Shared central-spin dependence prevents that simplification.

The owned exact checker evaluates (6.1) using only rational intervals: square-root bounds are certified by squaring integer-scaled endpoints, exponentials by positive Taylor sums with a geometric remainder, and logarithms by

$$\log x = 2 \sum_{k=0}^{n-1} \frac{z^{2k+1}}{2k+1} + \rho_n, \quad z = \frac{x-1}{x+1}, \quad 0 \le \rho_n \le \frac{2z^{2n+1}}{(2n+1)(1-z^2)} \quad (x \ge 1). \tag{6.3}$$

For $0 < x < 1$, use $\log x = -\log(1/x)$. This is an exact finite certificate for a specified capped hierarchy. It is neither an empirical error bar nor a test of thermodynamic equality. The resulting outward decimal enclosures, rounded much more coarsely than the saved rational endpoints, are

$$0.7254655628103138 < B_{\mathrm{cap}} < 0.7254655628103140,$$
$$0.9804601602442673 < B_{\mathrm{cap}} + e^{-1} \log 2 < 0.9804601602442675.$$

7 Source architecture and retained obligations

All eleven scientific modules of the cached manuscript were read literally, including the substantial perturbation and decorrelation proofs. Reading is not a formal verification of every limit, selection argument, or imported theorem. The following map states where the global argument lies and which features the local analysis must retain.

Source module	Mechanism and boundary retained here
`introduction`, `overview`, `model`	Counting prior, even arity, integrable interaction maximum and field, factorization/positivity, whole-chain trial labels, and an infimum over finite depths without attainment.
`trees`	General conditional probability kernels. Normalized insertion changes every ancestor, including unary vertices. Protected-anchor cancellation gives a logarithmic insertion series with a genuine remainder for $\|D\| < 1$. Its fixed-depth complex nonvanishing constants are not asserted uniform in depth. These identities are source results, not new local contributions.
`upper`	Poisson interpolation and the sign of $u^p - puv^{p-1} + (p-1)v^p$. First-moment extension regularizes an interaction ratio normalized at a spin configuration; it does not silently require $\mathbb{E}\|\log a\| < \infty$.
`cavity`	Bounded permutation-invariant reservoir, a new-site insertion with mean degree αp, and an old-bond correction with mean $\alpha(p-1)$. Repeated new-site indices and perturbation-count couplings have separate vanishing errors.
`perturbations`	A countable marked perturbation reservoir, normalized derivative identities, concentration, and deterministic parameter selection. The analytic coefficient argument takes $N \to \infty$ at fixed type/depth before sending the mark scale to zero.
`decorrelation`	Relative split/delay identities retain coefficients of order δ^{2b} with $\delta = 1/L$. The argument sums histories before taking absolute values and controls regular-depth *averages*. It does not provide concentration at every individual depth.
`comparison`	Nested conditional spin measures encode all needed multi-overlap patterns. Finite spin-pattern probabilities are Fourier expansions in all subsets, not functions of pair overlaps alone. Count and series cutoffs precede the N, L limits.
`reduction`	Integrable common-shift removal, bounded-law approximation, and uniform trial/pressure comparison restore the original first-moment model. Entry clipping need not retain the original factorization.
`consequences`	Symmetric-coupling even-p models, even-arity soft constraints, and a ground-state limit with a separate entropy error $\log 2/\beta$. Odd-arity random 3-SAT is explicitly outside this theorem.

Two interfaces deserve particular care. In the source comparison, conditional on a finite replica tree, the empirical site-pattern law is

$$\pi(e) = 2^{-k} \sum_{S \subseteq [k]} R_S \prod_{a \in S} e_a.$$

Uniformly sampled sites with replacement yield the appropriate product law *conditional on that tree*. Replacing R_S by $\mathbb{E}R_S$ therefore requires all relevant subset errors, not just a matched one-site or covariance law. The depth averages in the source provide that ingredient only after their own limits.

For the series comparison, after a count cutoff M and bounded disorder B, H, the source uses an insertion range controlled by $C = H + BM$ and $c = \tanh C < 1$. Its log-series tail is bounded by $\sum_{k>q} c^k/k$. The count cutoff, series order, fixed-depth thermodynamic limit, depth limit, and

regular-depth exclusion must be taken in the stated order. Our degree cap bounds an evaluated trial directly; it does not remove these obligations or turn the source argument into a uniform rate in N and r.

No source correction was established in this study. Important valid safeguards retained are the quenched physical variables, independent whole label chains, normalization derivatives, retained unary paths, regular-depth averaging, and the separation of upper-bound positivity from bounded-law lower-bound reduction. The local theorems are proved without importing the source's global decorrelation claim.

8 Original Ram connection and evidence scope

The original local-moment proof [8], lines 26–63, expands an integral with a finite Taylor remainder and an absolute-moment majorant. Its normalization discussion distinguishes exact preservation of constants from positivity, while lines 194–225 distinguish polynomial exactness from closure or invertibility. Applied here, the kernel is a *conditional probability law of a continuation*, the tested function is ϕ_m, and the remainder in Theorem 5.1 is explicit. No differential-operator inverse, finite-rank closure, or reconstruction of arbitrary laws is invoked. The chapter's tail separation at lines 314–330 also motivates keeping physical truncation outside the conditional polynomial approximation.

The original normalized-kernel calculation in [9], lines 399–418, derives a value derivative plus a centered normalization term; its quotient identities at lines 2104–2116 make the same gate explicit at second order. These are the actual proof passages used in (5.4), not an inference from a catalogue title. The later residual/cubature report [10], lines 436–457, supplies the finite polynomial-error plus $(1 + \Lambda)$ remainder bound. Its independent-final-sample discussion was also read, but no statistical unbiasedness or sampling confidence claim is used here. In particular, an empirical collection of moment estimates is not declared an exact positive cubature.

The fresh literal ledger covers all 15 public text files: the 11 modules, main file, glyph mappings, bibliography, and README, totalling 3,058 lines. The private coverage is 948 distinct lines: the entire 446-line local-moment chapter; lines 1–98, 328–426, 2104–2141 of the 2,250-line nonlinear report; and lines 1–140, 165–267, 436–459 of the 768-line residual report. The chapter's six input files and all other private files were not read for this report. Their contents are not implicitly included by reading an `input` command. The ledger records raw displayed intervals, merged intervals, unread complements, and before/after SHA256 hashes. Captured repository identity is provenance, not a live branch or proof-validity assertion.

The independent owned checker passes 940 parameter cases with 2,868 counted assertions. It tests finite total-variation inequalities, product couplings, midpoint offsets, Poisson identities with their finite-sum endpoint, conditional moment budgets, and the explicit trial. It uses exact fractions and rational enclosures, not floating-point acceptance thresholds. Its machine receipt distinguishes cases from assertions; floating-point numbers are display-only approximations of already certified endpoints. None of these fixtures checks the source perturbation identities at all orders, the depth limit, or variational optimality. External review checks are not counted as author-run tests.

9 Public status, advantages, and remaining work

The October 9, 2026 primary-source refresh separates several claims. Mézard–Parisi [2] proposes a one-step cavity description on the Bethe lattice; this is not a proof of the present all-depth equality.

Panchenko–Talagrand [3] supplies the classical interpolation-bound framework. The sign convention for a free energy can differ from the positive pressure (1.2). Panchenko's spin-distribution formulation [4] describes invariance-based order parameters. His finite-RSB result [5] identifies a perturbed finite-RSB spin structure and explicitly distinguishes the missing general full-RSB approximation in that work. These are historical mathematical inputs, not a statement that an abstract alone proves the cached manuscript.

The more recent replica-symmetric result of Biswas–Chen–Sen [6] applies in specified high-temperature and subcritical regimes, the latter involving $\alpha p(p-1) \leq 1$; it is not a replacement for a general all-temperature hierarchy formula. Concetti [7] studies a full-RSB functional on random regular graphs through a stochastic formulation. That graph model and functional should not be silently substituted for the Poisson trial here. These refreshed abstracts and publisher scope statements were used for status, not as full proof audits. The supplied OpenAI manuscript is a September 23 preprint with a claimed equality. This report neither promotes it to an independently established settlement nor claims the public search proves absence of other work.

The practical advantages of our local refinements are specific. The message error depends on spin contrast rather than a common energy offset, is uniform over all finite hierarchy depths, and vanishes under the original first moment. The degree cap pays only for omitted interactions. The conditional evaluator preserves normalization and has a regular small-mass limit. The combined budget (4.4) states exactly how an approximate trial can retain a valid upper-bound interpretation.

The costs are equally concrete. Tail stop-loss integrals and offset means need certified input. The contrast bound may be conservative for interactions with additional structure. Very broad continuation ranges can make Taylor evaluation expensive. Finite-alphabet messages do not control branching complexity or approximation of a law on laws. Signed rules lose automatic positivity and can amplify errors. Finally, finding a trial with small value remains a nonlinear optimization problem; a numerical stationary point does not prove global minimality.

Useful further work is to exploit spin-specific contrast ranges for each cavity message, build adaptive positive rules for conditional continuation laws, and allocate certified errors according to the actual site and edge costs. A genuinely finite-branching approximation with quantitative control of all conditional kernels would extend Corollary 3.2; it has not been proved here. Relating such finite evaluation to a two-sided thermodynamic estimate would additionally require an independently controlled lower-bound mechanism, not merely more quadrature nodes.

10 Conclusion

The new local content is the uniform spin-contrast approximation, its finite-alphabet infimum consequence, and the assembled first-moment certificate coupling message, degree, physical, and conditional evaluation errors. The normalization-aware evaluator provides a compatible way to use the original Ram finite-remainder method without introducing false exponential-moment assumptions or a false small-mass singularity. Classical coupling, log-sum inequalities, Poisson summation, Taylor remainders, and quotient differentiation are credited as the tools, not relabeled as discoveries. C3 reflects the demonstrated source-specific local increment, independently of L3 applicability. The global variational formula, its optimal law, and an effective all-depth optimization rate remain outside the certificate.

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